



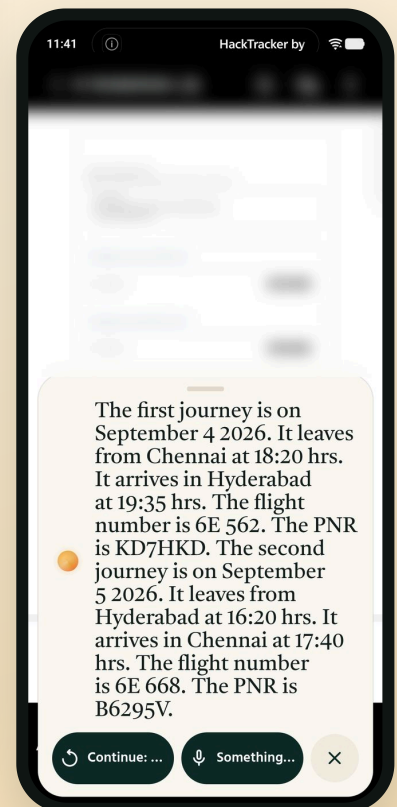
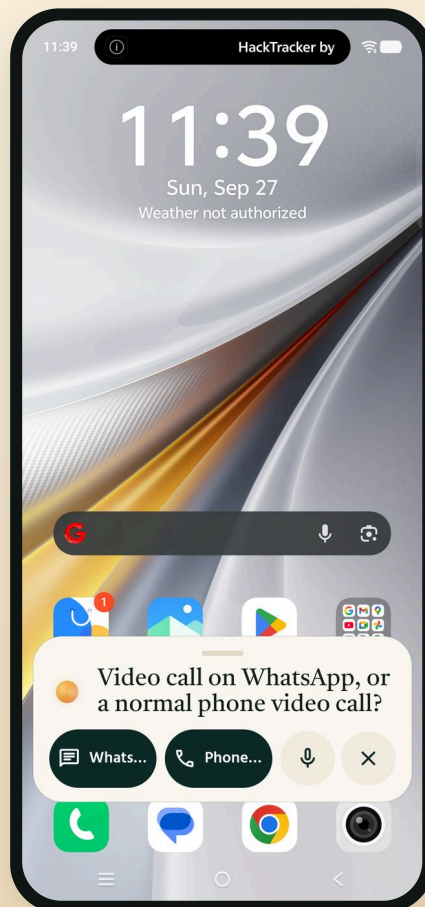
An on-device AI companion that guides India's elderly through **their own phone** — in English, हिंदी and తెలుగు — by glowing exactly where to tap, saying one simple step, and doing it with them, safely.

Runs offline on the iQOO 15

NPU + GPU + CPU, every engine used

3 languages, spoken

Never touches money, OTPs or IDs



A companion that sits on top of every app

The person says what they want, the way they would ask a grandchild — “*video call my son*”, “*find my Aadhaar and send it to him*”, “*Aakash sent me a flight ticket, when is it?*” — and Saathi walks them through it on the real app: a warm glow around the exact button, one short sentence in their language, spoken aloud, and a **Do it for me** button when they just want it done.

02 · PROBLEM STATEMENT

Digital India left its elders one tap behind

Pensions, Aadhaar, train and flight tickets, bills, doctors, family — it all lives in smartphone apps now. Yet millions of parents and grandparents were never taught how these apps work, many read their own language far better than English, and every phone brand and every update moves things around.

Every task needs someone else

A video call, sending a photo, bigger text, finding a document — each waits for a son, daughter or neighbour to be free. Asking again and again costs independence and dignity.

English-first, ever-changing apps

Menus and icons change with every update and brand. A step learned last month is somewhere else today. Almost all help is English and generic.

Fear of breaking something

One wrong tap can call a stranger, delete a chat or open a popup nobody can close — so many simply stop trying.

Scams aim at exactly this person

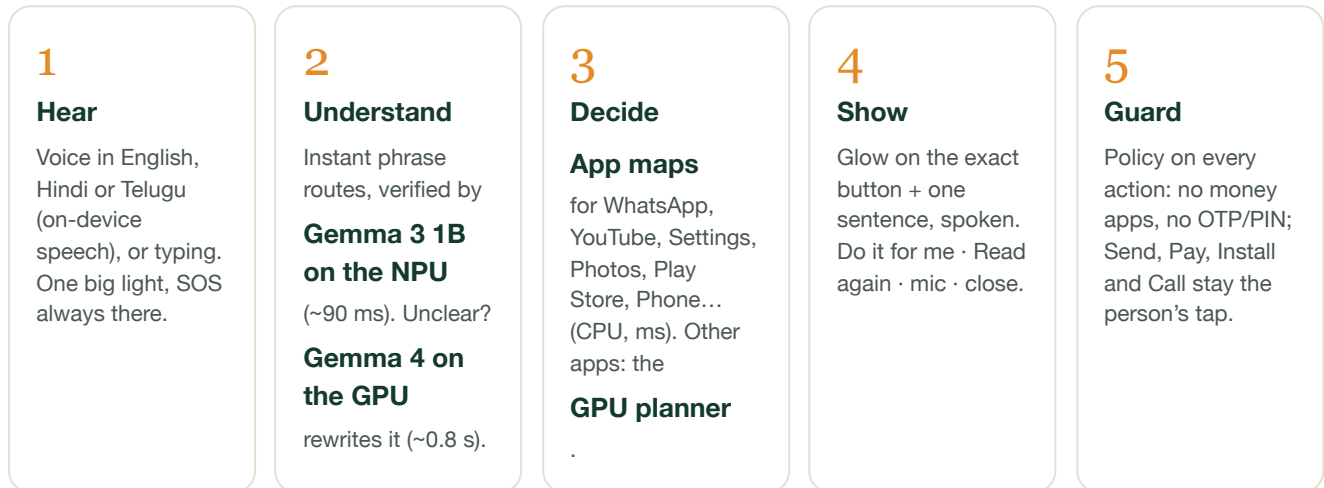
“Share your screen”, “install this app”, “tell me the OTP”. Those least sure of their phone are the easiest to trick, and remote “help” is how most of these frauds begin.

Why today's tools don't close the gap

- **Voice assistants** run a handful of commands and answer questions — they don't walk you through the app on your screen, don't teach, and mostly need the cloud.
- **Screen readers** are built for blind users; they read everything aloud but don't know your goal.
- **Remote help** from family needs someone free and online, and trains people to let others control their phone — the habit scammers exploit.
- **Video tutorials** are generic, English, for another phone, and you can't ask them anything.

Understand the goal. Show the next step. Keep them safe.

Saathi is an Android **accessibility-service overlay** with a stack of **on-device models**. It reads the structure of whatever app is on screen (the same accessibility tree screen readers use), understands what the person wants, decides the next step and shows it on the real app — then watches the screen change and moves on.



Key features

- **Guides on the real app** — glows the actual button, recovers when the person wanders (“I’m lost” → where you are and the way back).
- **Teach me, or Do it for me** — same task, two modes; the final risky tap is always theirs.
- **Three languages, mixed freely** — speech, cards and answers; Telugu/Hindi names become English letters to find contacts (అకాష్ → akash).
- **Understands natural sentences** — the NPU checks every route, the GPU rewrites unclear requests and plans in apps nobody mapped.
- **Reads documents** — screen capture → OCR → Gemma 4: tickets, letters, forms, labels.
- **Writes for them** — letters and replies composed on-device, typed into the box they are looking at.
- **Never loses the thread** — questions mid-task are answered while the task waits; interrupted tasks come back with “Continue”.
- **Scam guard** — warnings on scam SMS, APKs, remote-control apps, money apps opened during a call.
- **Live AI monitor** — shows which engine ran each step and how long it took.
- **Saathi Pro** — a separate optional app with cloud help for power users; basic Saathi stays fully offline.

~90 ms

NPU check on every request

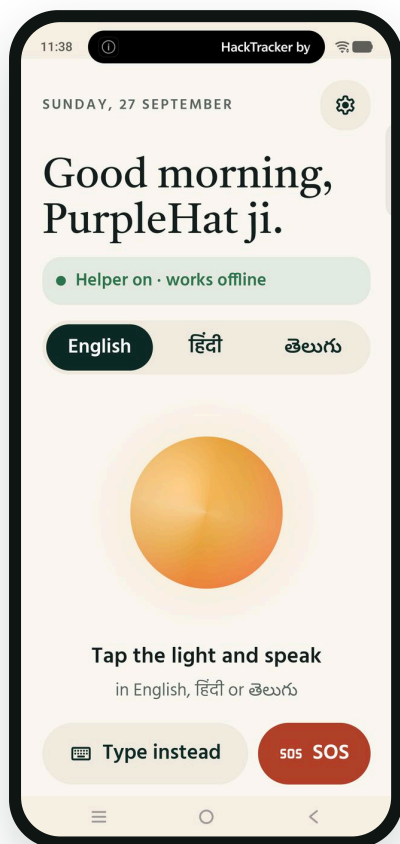
0.8–2.5 s

GPU: understand, plan, write

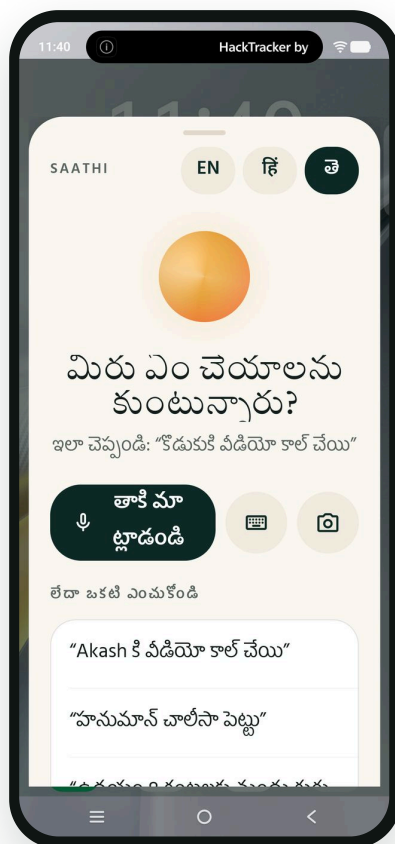
0 bytes

leave the phone — basic Saathi has no internet permission

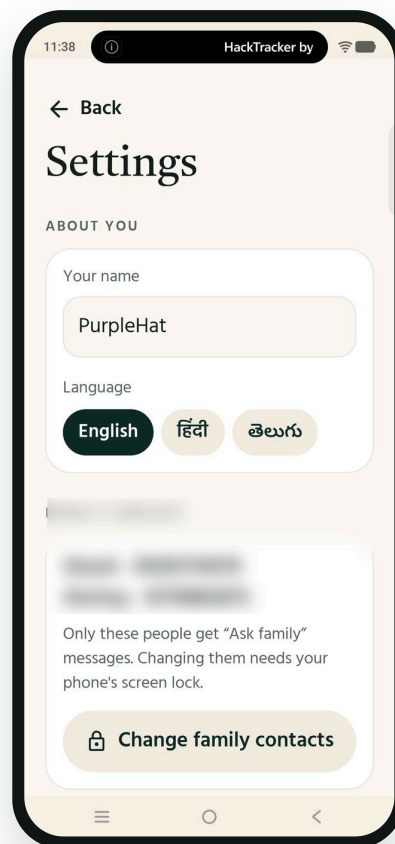
Calm, large, and in their language



Home. One light to tap and speak, three languages, Type instead, SOS. “Helper on · works offline”.



Listening in Telugu. Examples in their own language: “call Akash on video”, “play Hanuman Chalisa”.



Settings. Name, language, family contacts — changeable only with the phone’s own screen lock.

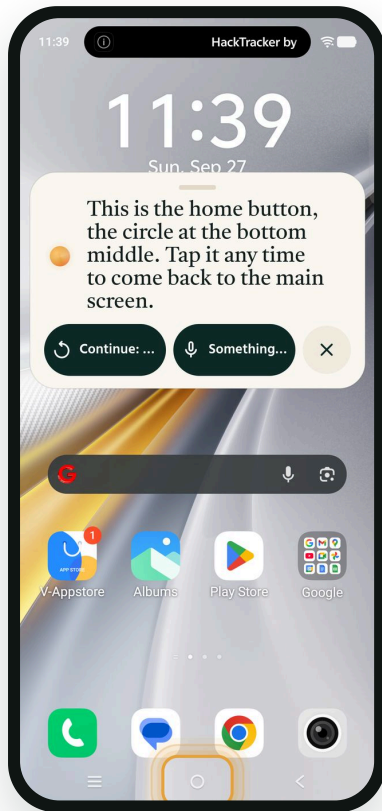
Designed for the person, not the phone

- Warm paper colours, large serif text, one action per card.
- The card never covers the button it points at, nor the keyboard.
- Text size, voice speed, dim-around-the-glow and scam warnings in Settings.

Two apps, one switch

- **Saathi:** fully on-device, no internet permission.
- **Saathi Pro:** adds a cloud model (the user’s own OpenRouter key) for power-user tasks such as SSH or video editing.
- “Use this Saathi” makes one active; the other steps aside.

Every use case, step by step



01

“Where is the home button?”

“This is the home button, the circle at the bottom middle. Tap it any time to come back to the main screen.”

HAPPENS

The real Home button on the navigation bar glows. Also back and recent apps, in all three languages — instantly, no model needed.

SOLVES

The most basic fear of all: “how do I get out of here?”

UNIQUE

It points at the phone itself, not at a drawing of a phone.

CPU · instant



02

“Video call my son”

“Video call on WhatsApp, or a normal phone video call?”

HAPPENS

“My son” is the family contact set up once. One plain question, then WhatsApp → search → his name → the video-call button in his chat glows. A wrong chat is noticed (“tap the back arrow”). The call itself is their tap.

SOLVES

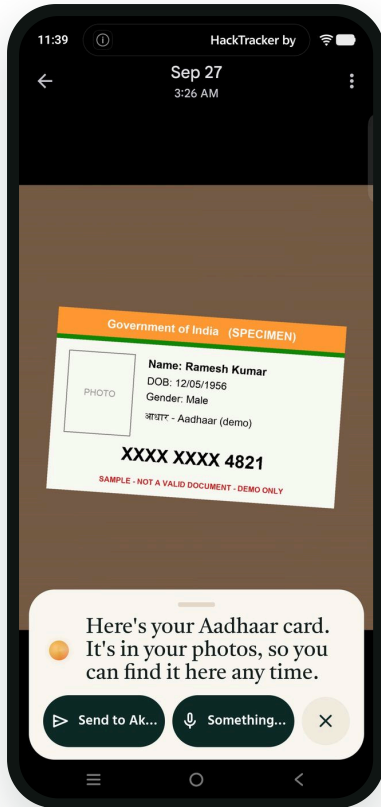
The number-one reason an elder hands their phone to someone else.

UNIQUE

Same in Hindi and Telugu — “बेटे को वीडियो कॉल करो”, “కొడుకు వీడియో కాల్ చేయి”.

NPU · verify route

CPU · WhatsApp map



03 · Highlight

“Find my Aadhaar” → “send it to my son”

“Here’s your Aadhaar card. It’s in your photos, so you can find it here any time.” [Send to Akash]

HAPPENS

On the phone, Saathi has already read the photo gallery with ML Kit OCR and remembers only “photo #123 is an Aadhaar” — never the number. It opens that picture. “Send it to my son” opens WhatsApp’s share sheet with it attached; “it” is understood from context. Works for PAN, voter ID, licence, prescriptions and bills. (Shown: a SAMPLE card.)

SOLVES

“Which of 4,000 photos was the Aadhaar?” — asked at every office, hospital and bank counter.

UNIQUE

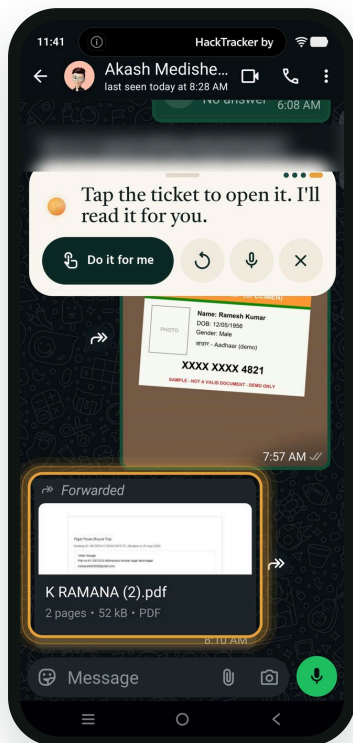
Private by design: no document leaves the phone, no numbers are stored, Saathi never types an ID number.

CPU · ML Kit OCR index

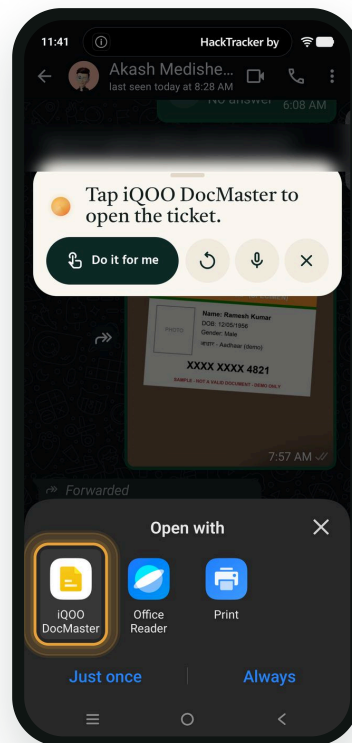
NPU · verify

04 · Highlight

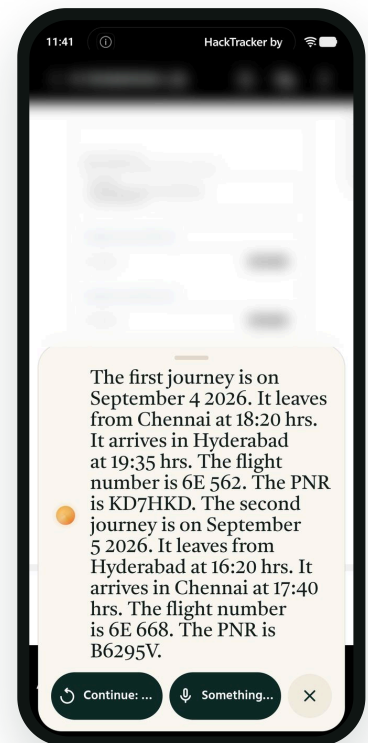
“Akash sent me a flight ticket on WhatsApp — when is it, from where to where?”



1 · His chat — the newest document glows



2 · “Open with” — the app glows, then Just once, Open only



3 · Read aloud: both journeys (personal details blurred)

Happens

Screen capture → ML Kit OCR (~0.3 s) → text rebuilt row by row so every city keeps its own time → Gemma 4 on the GPU (~7 s): “4 Sep 2026, Chennai 18:20 → Hyderabad 19:35, 6E 562 ...”. In Telugu the answer comes in Telugu.

Solves

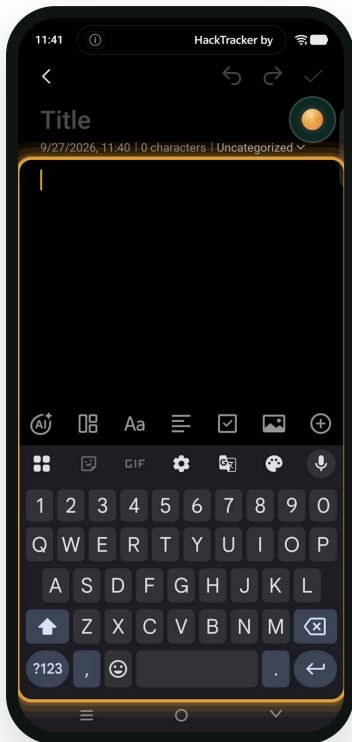
Tickets, reports and bills arrive as PDFs in tiny print, usually in English. Now they can simply ask.

Unique

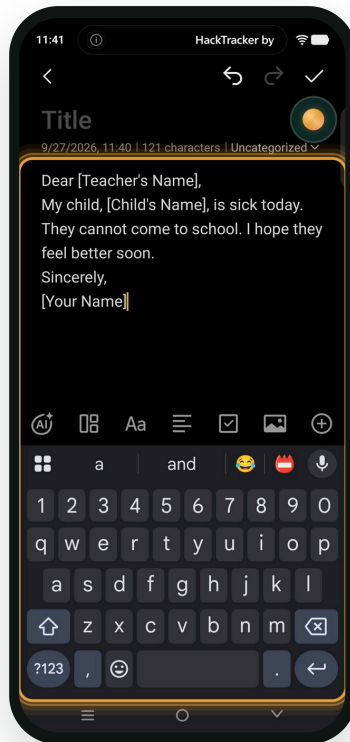
End to end on the phone — the booking is never sent to a server — and the answer is checked journey by journey, never mixed.

05

In Notes: “Write a sick-leave letter for my kid”



An empty note — Saathi knows where they are



Gemma 4 composes, “Do it” types it into the body

HAPPENS

On-screen context: the note body is the box. Gemma 4 writes a proper short letter on the GPU (~2.4 s) with greeting and closing. Misheard speech (“writer a letter”) still works.

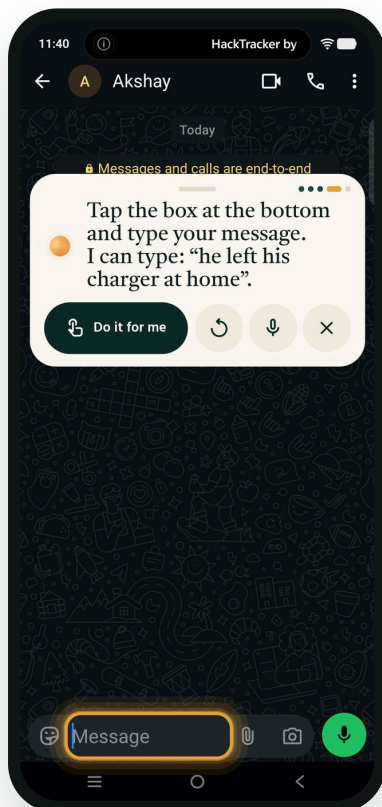
SOLVES

Writing is the hardest part of a phone for many elders.

UNIQUE

Writes where you are; never turns into plain dictation.

GPU · compose



06

“Send a message to Akshay that he left his charger at home”

“Tap the box at the bottom and type your message. I can type: ‘he left his charger at home’.”

HAPPENS

The NPU doubts the first guess; Gemma 4 rewrites it as a WhatsApp message to Akshay; the map finds him and fills the words. Sending is their tap. Also **voice notes** (“press and hold the mic, speak, let go”) and **“take a photo of this paper and send it to my son”** (WhatsApp’s in-chat camera).

SOLVES

Everyday messages without hunting through menus.

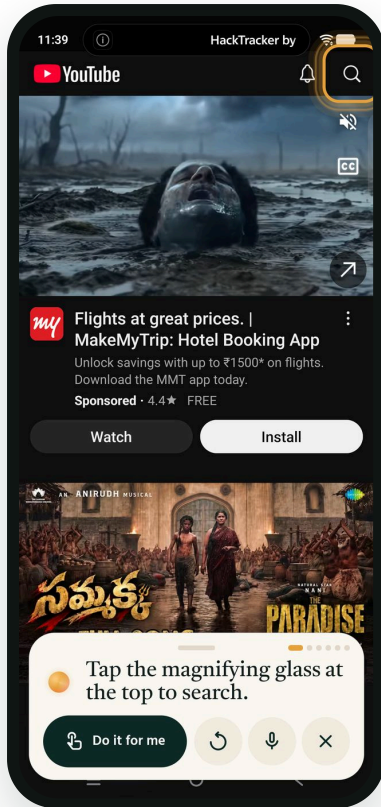
UNIQUE

Natural sentences, not commands; contact names work from Telugu and Hindi too.

NPU · verify

GPU · rewrite

CPU · WhatsApp map



07

“Play Hanuman Chalisa on YouTube” — and control it mid-video

“Tap the magnifying glass at the top to search.”

HAPPENS

Search → type → the first real video (not an ad or a Short) → playing. While it plays, “pause”, “louder” or “close it” just work and the task keeps its context. Also liked videos, watch later, downloads, “play all” playlists and JioHotstar titles. Ads, “rate us” and notification popups are handled first: “A popup is in the way — tap Dismiss.”

SOLVES

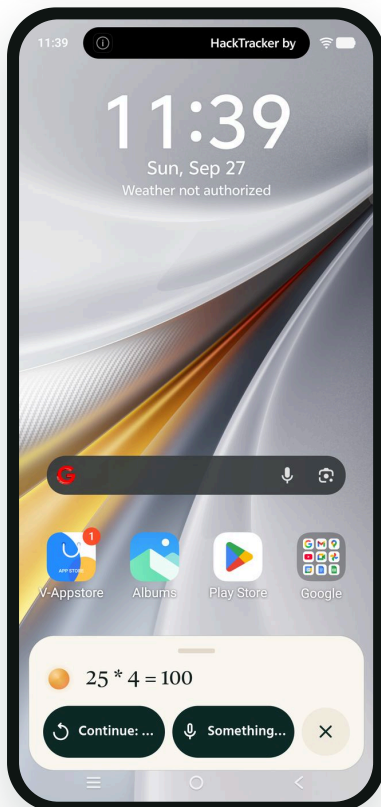
Devotional songs, news and serials — the everyday reasons elders pick up a phone.

UNIQUE

Follow-ups belong to the same task instead of starting over.

NPU · verify

CPU · YouTube map



08

Quick answers — and “Continue” where you left off

“ $25 \times 4 = 100$ ” [Continue: send a WhatsApp message...] [Something else]

HAPPENS

Sums are instant; questions are answered by Gemma 4. If a task was cut short — a new request, leaving the app — the next card offers **Continue**, or they simply say “continue”. Torch, volume and alarms mid-task just happen and the task carries on. Models load at task start and stay warm for 15 minutes.

SOLVES

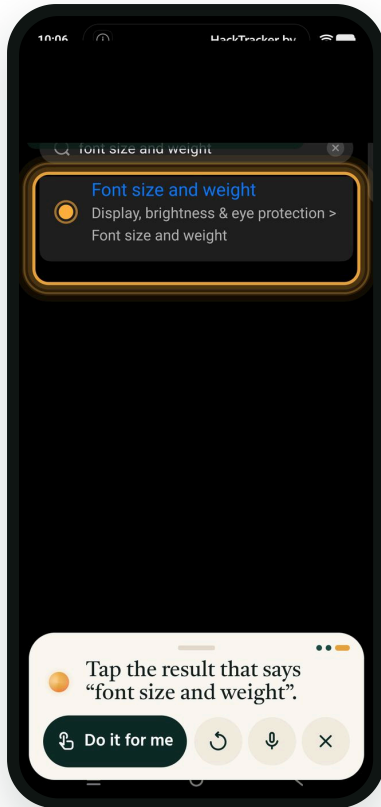
Real life interrupts; a helper that forgets is no helper.

UNIQUE

The companion remembers where you were.

CPU · instant

GPU · answer



09

Phone settings — “make the letters bigger”, battery percentage, brightness, ringtone, Wi-Fi

“Tap the result that says ‘Font size and weight’.” ... “Is it good like this?”

- HAPPENS** Uses the phone’s own Settings search, so it works on every brand: types the exact term, glows the right result (never another app’s setting), then watches the slider or switch live and asks “Is it good like this?”. vivo switches helpers off when Settings opens — Saathi restores itself in a fraction of a second and continues.
- SOLVES** “The text is too small” — the most common complaint of all.
- UNIQUE** It finishes the last mile — the slider — instead of just opening a page.

Apps Saathi already knows (57 verified routes)

WhatsApp video/voice call, message, photo, voice note, take-a-photo-and-send, open a document, see a photo, backup, share location · **YouTube** search, subscriptions, history, like, share · **Google Maps** directions & navigation, **book a cab** (Maps → Uber) · **Uber** cab booking · **Photos** open latest, crop, brightness, share · **Phone** call · **Messages** read latest, block · **Clock** new alarm, alarm off · **Camera** photo, selfie · **Chrome** search, new tab · **Instagram** profile, reels, post · **Play Store** install, update all, uninstall · **Spotify** play, playlist, like, shuffle, download · **JioHotstar** watch · **Google Docs** new, open recent, save as Word · **Settings** ringtone, font, brightness, wallpaper, Wi-Fi, Bluetooth, dark mode, language, screen timeout, storage.

Plus 31 instant skills (torch, volume, battery, TV remote, alarms, medicine...) and the GPU planner for any other app.

Paper form helper — “help me fill this form”

Aadhaar enrolment form: field 1 of 8, the box glows, the value in big letters to copy

Not in the saved details (“Post Office”): “ask your family”

Bank form: “To be filled in by the bank” is left out; 5 fields for the person

Happens

Point the camera at a paper form — or share a photo of it from the gallery or WhatsApp (“Fill this form with Saathi”). ML Kit OCR on the phone finds each label and its box, then walks one field at a time: zoomed to the box, the box glows, the card says what to write and shows it in big letters. Back / Next / Read again. Saathi never writes on the paper and nothing leaves the phone.

Leaves out what isn't theirs

Sections meant for someone else are skipped by position and heading: “To be filled in by the bank”, “Certifier: to be filled by the District Child Protection Officer”, “For office use”, signatures, stamps, photo boxes. Aadhaar, PAN, bank and card boxes are never filled — the person writes those.

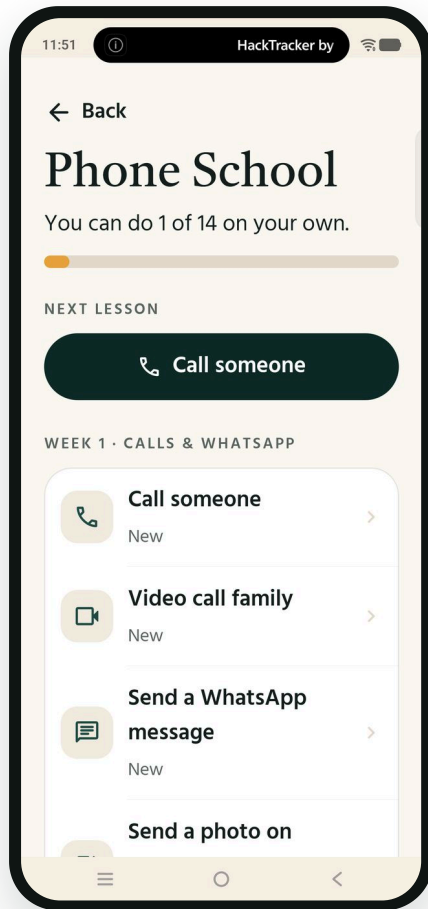
Values come from “My details”

The answers are preset once in the app's **My details** profile (opens behind the phone's screen lock). In this demo the profile holds sample values — name “Akash”, house/street “None”, area “Neon”, city and district “Karimnagar” — “None” is itself a preset value, not an OCR mistake. Missing ones (middle name, post office) are said plainly, never guessed.

Online forms too

On any form inside an app or web page, “help me fill this form” walks the editable fields in reading order, glows each and offers to type the saved value. Password, OTP and PIN fields are recognised and never filled.

Saathi learns with the person — and from the family



Phone School. Every skill as a small lesson; progress comes from doing it.



“Watch me ...” A family member shows a task once.

Fading help

Each skill is tracked: new → practising → “You can do this!”. After three successes Saathi lets the person try first and only glows if they get stuck — teaching, not replacing.

Teach once — “watch me”

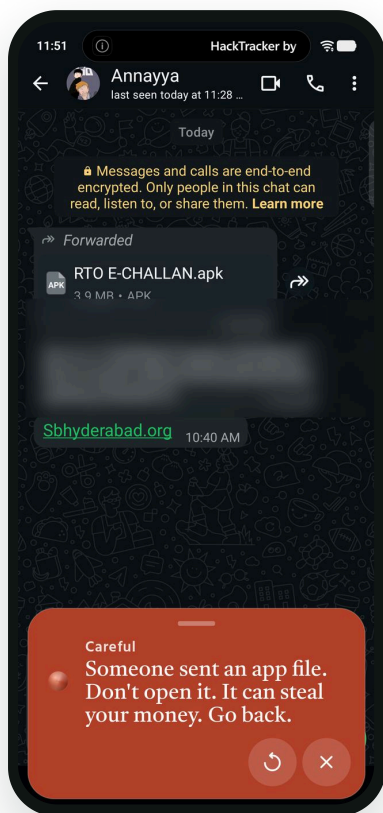
Saathi records the taps and safe typed values of a task shown once (“watch me open the school app”) and saves it as a lesson, replayed later with glows, step by step, by name and by voice.

New apps & new things

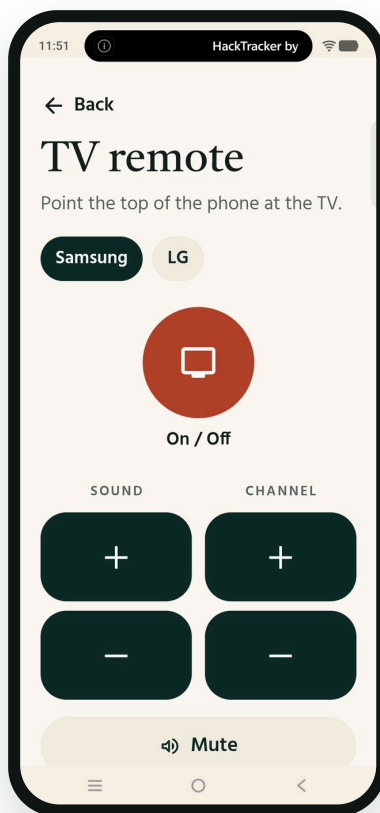
“How do I use Spotify / Google Docs?” gives guided first steps on the real app. For apps no one mapped, Gemma 4 plans each step from what is on screen, with drift checks (“where am I?”).

How the “model learns”: models are not retrained on the phone. Saathi learns by storing lessons, labels and progress in private on-device memory; the models receive that memory as context, so the same Gemma plans better on this person’s phone the more it is used.

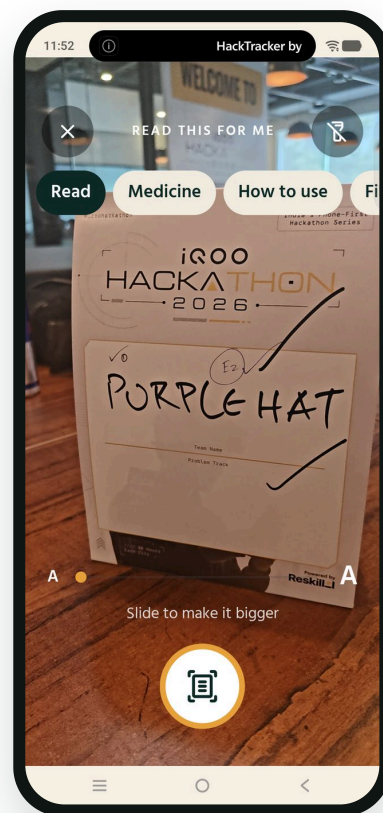
Small things that matter every day



Scam guard, caught live: an “RTO E-CHALLAN.apk” forwarded in a chat → “Don’t open it. It can steal your money.”



TV remote through the iQOO’s IR blaster — huge buttons, or by voice: “TV volume up”, “channel 245”.



Read this for me — magnifier + OCR; letters, bills, signs; medicine strip → reminder.

SOS

“Help!”, “बचाओ”, “కాపాడండి” → one screen: call family, call 112, prefill “I need help”.

Ask family — can’t be hijacked

Prefills a message to the registered family contact only; redacted; the person presses send; no remote control ever.

Routines & reminders

“Every morning at 7 play Hanuman Chalisa”, medicine reminders — spoken, confirmed with Yes, re-armed after reboot. Battery-aware: under 15% it skips the big model.

Add a skill once — it works everywhere

In

Any way of asking

Voice or typing,
English/Hindi/Telugu,
mixed; follow-ups mid-
task; misheard words.



One meaning

Phrase index over every
skill → NPU verifies →
GPU rewrites
paraphrases onto a
known skill; slots
(contact, text, place,
query) extracted;
Telugu/Hindi names
transliterated.



Knowledge

31 skills

.

57 app routes

in 16 apps ·

family lessons

(watch me) ·

Gemma 4 planner

for everything else.



Same runtime

Glow + card + voice, Do
it for me, last-mile settle,
Continue, safety policy,
AI monitor.

What “adding a skill” means

- **A skill** = keywords + steps (or an instant action) in one registry.
- **An app route** = screens described by what is on them (labels, ids, positions) + steps with slots — one file per app, tested against real captured screen trees.
- **A lesson** = recorded once by the family, no code.

What it gets for free

- Works by voice in three languages and with any phrasing (the models map paraphrases onto it).
- Resumes after interruptions; answers doubts mid-task; recovers when lost.
- Every safety rule applies automatically; unknown screens fall back to the planner.

Every engine on the iQOO 15 has a job

A small live monitor (switchable in Settings) shows which engine ran each step. These lines are real measurements from the phone:

SAATHI · ON-DEVICE — AI MONITOR

NPU	Gemma 3 1B	verify route	91 ms
GPU	Gemma 4 E4B	rewrite request	789 ms
CPU	App map	WhatsApp · step 3	5 ms
GPU	Gemma 4 E4B	compose letter	2 403 ms
CPU	ML Kit OCR	read ticket	239 ms
GPU	Gemma 4 E4B	read ticket	7 600 ms

Engine	Model	What it does in Saathi	Time on the phone
NPU	Gemma 3 1B · LiteRT-LM	Double-checks every quickly matched request; fast intent reading	~90 ms
GPU	Gemma 4 E4B · LiteRT-LM	Rewrites unclear requests, plans steps in unmapped apps, writes letters and replies, answers questions, reads tickets and papers	0.8 s – 7 s
NPU	FastVLM	Camera: “what is this?”	on demand
CPU	ML Kit OCR (Latin + Devanagari)	Screens, documents, the private gallery index, paper forms	0.2 – 0.5 s
CPU	App maps	Verified screen models for WhatsApp, YouTube, Settings, Photos, Play Store, Phone, Maps, Spotify, JioHotstar...	2 – 20 ms / step
Cloud	OpenRouter (Saathi Pro only)	Picks the right app and plans steps for power-user tasks	~2.5 s / step

Models load when a task starts and stay warm for 15 minutes after the last use, then give the memory and battery back.

Not an assistant that acts for you — a companion that goes with you

	Voice assistants	Screen readers	Remote help	Saathi
Guides on your real app	A few commands	Reads everything aloud	Someone else taps	Glows the exact button, step by step
Teaches, or does it	Does, sometimes	—	Does	Both: “teach me” or “do it for me”
Keeps the goal across apps	Partly	No	Yes (a person)	Yes, and resumes after interruptions
Offline & private	Mostly cloud	Yes	Needs internet	Fully on-device; no internet permission
Hindi & Telugu	Partly	Yes	Depends	Speech, cards, names and answers
Scam-safe	—	—	Trains people to allow control	Refuses money/OTP, warns in the moment

08 · SAFETY & PRIVACY — ENFORCED IN CODE ON EVERY ACTION

- Never types or stores passwords, OTP, PIN, card, CVV, Aadhaar, PAN or bank numbers.
- Never presses Send, Pay, Install or Call — the final tap is always the person's.
- Never guides inside banking or UPI apps: “please do this yourself, or with someone you trust.”

- Basic Saathi has no internet permission; the document index keeps only “photo → kind”.
- Saathi Pro is a separate app; the key is the user's own; personal data is redacted before any request.
- Family contacts can only be changed with the phone's screen lock.

“The banks are right to block unknown screen readers — that's how elders get robbed. Saathi never works inside money apps; it protects the person on the way in.”

Engineering

1,150+

JVM unit tests — routing, app maps checked against real captured screen trees, forms, safety, languages

13

scripted end-to-end scenarios run on the iQOO 15 in English, Hindi and Telugu

2 apps

Saathi (offline) and Saathi Pro (cloud-assisted), one switch

- **Stack** — Kotlin; Android AccessibilityService with overlay windows; LiteRT-LM running Gemma 3 1B (NPU) and Gemma 4 E4B (GPU); FastVLM; ML Kit text recognition; on-device Android speech; OpenRouter for Saathi Pro only.
- **App maps** — each screen of each app is described by what is on it (labels, ids, positions); every step is tested against captured screen trees, so a changed app breaks loudly in CI, not silently for a grandmother.
- **Resilience** — vivo switches accessibility helpers off when Settings opens; Saathi restores itself and resumes the task. Only one Saathi app is active at a time.

10 · WHAT'S NEXT

From one phone to every elder

Bank and secure apps

Saathi steps aside so the bank app opens, briefs the person before and checks on them after. Long term, a preinstalled Saathi becomes a trusted accessibility tool — the same tier as TalkBack.

More of India

Tamil, Kannada, Marathi and Bengali; more app maps; a family dashboard to share lessons across a household.

Saathi gives an elder back the one thing a phone quietly took away: doing it themselves.